Time Series Change Vector Analysis for Semisupervised Abrupt Land Cover Change Detection

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***Abstract*—Change detection (CD) in satellite image time series (SITS) is more complex than in bitemporal images due to the higher dimensionality of the data. Utilizing the full dimensionality of the time series remains challenging, particularly with dense SITS. An approach that can minimize dimensions without compromising informational depth is essential. In this article, we present an innovative framework for change vector analysis (CVA) in time series analysis and initial demonstrations of its effectiveness in capturing the spectral–temporal characteristics of changes. Unlike current methods, the proposed approach incorporates a wide range of spectral–temporal information and constructs separate reference matrices for each change type, facilitating an in-depth analysis of change components for CD. Based on the time series change vector (TSCV), the proposed framework extends CVA into the time series perspective, offering novel interpretations for magnitude and direction across temporal and spectral dimensions. The framework's effectiveness is validated using Sentinel-2 data, demonstrating significant improvements in tackling multiple CD challenges in dense SITS scenarios.**

***Index Terms*—Change detection (CD), change vector analysis (CVA), multispectral, multitemporal, remote sensing, satellite image time series (SITS), semisupervised, time series CVA (TSCVA).**

I. Introduction

SATELLITE remote sensing facilitates the continuous monitoring of land cover changes over extensive geographical areas, surpassing the traditional in situ approach, which necessitates significant time and resource investments [1], [2], [3]. Present-day Earth observation satellites gather vast volumes of data that are rich in spectral and temporal information, proving valuable for detecting land cover changes [4]. Changes in land cover are among the critical environmental issues that demand continual focus. Such dedication contributes significantly to formulating policies to foster a sustainable future for our planet [5]. Change detection (CD) is developed for recognizing changes that have taken place in a specific geographic area over time [1], [3], [6]. In remote sensing, CD approaches have been applied in various fields, including forestry [3], ecosystem monitoring [5], agriculture [7], urban planning [8], and disaster management [9].

CD techniques in satellite image time series (SITS) remain underrepresented in the literature. One such method, the post-classification comparison (PCC) approach, has been employed in time series analysis [10], [11]. This technique involves performing separate supervised classifications on images of the same geographic area, taken at different times, and then comparing the resultant maps to evaluate land cover changes [3]. Nonetheless, PCC does not consider the temporal correlation between the images [12] and heavily depends on the accuracy of the classification algorithms, which significantly impacts its performance [13]. Another approach, namely the continuous CD and classification (CCDC) algorithm, decomposes SITS into its constituent seasonal, gradual, and residual elements, employing a model where coefficient values are forecast through the utilization of the ordinary least squares (OLSs) technique [14], [15]. Subsequently, the projected model is juxtaposed against cloud-free observations for the identification of changes, employing band-specific thresholds as criteria. The breaks for the additive seasonal and trend (BFAST) algorithm, on the other hand, can detect and differentiate between changes within the trend, seasonal, and remainder components of SITS [16], [17]. BFAST iteratively determines both the temporal occurrence and quantity of changes, describing each change based on its magnitude and direction. More recently, the Bayesian estimator of abrupt change, seasonal change, and trend (BEAST) has been proposed as an ensemble-based time-series decomposition algorithm. BEAST improves robustness by averaging across multiple competing models rather than relying on a single "best" one, enabling reliable detection of changepoints, seasonality, and nonlinear trends, while also providing credible uncertainty estimates [18]. However, these methods still require a priori information in the form of reference labels. The availability of reference labels is rarely assured and, in the context of time series, is effectively nonexistent. Recent CD studies predominantly utilize deep learning methods, which necessitate extensive training samples [19], [20], and do not explicitly address the spectral–temporal dimension. Our proposed framework differentiates itself by integrating spectral, temporal, and combined spectral–temporal

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analyses, thus providing a more comprehensive approach. Consequently, research efforts have shifted toward unsupervised techniques that do not rely on labeled data. However, unsupervised CD approaches for SITS are still relatively underexplored in the literature [21].

Within the realm of CD, change vector analysis (CVA) is a renowned approach employed to distinguish between various change types [22], [23], [24], [25]. This differentiation is achieved through the computation of spectral change vectors (SCVs), which highlight the discrepancies in temporal trajectories between prechange and postchange. Compared to other time-series approaches such as CCDC, BFAST, and BEAST, CVA-based methods differ fundamentally in their operating principle. CCDC and BFAST model each pixel's temporal trajectory by decomposing it into seasonal, trend, and residual components, detecting changes through parametric fitting and break identification, while BEAST uses a Bayesian framework to jointly estimate change timing, magnitude, and uncertainty. These model-driven approaches are effective for dense, regular time series but require parameter tuning, assumptions about temporal structure, and may be sensitive to noise or irregular acquisitions. In contrast, CVA operates directly in the spectral domain by computing SCVs between acquisitions, capturing both magnitude and direction of change without fitting temporal models. This reduces computational complexity, removes the need for seasonal–trend assumptions, and makes CVA well-suited for irregular time series or limited historical archives. SCVs are differentiable into either change or no-change groups, while also offering insights into the particular nature or type of the change [24]. CVA-based CD methods evaluate changes in SCVs across multiple dimensions and also benefit from having a comparatively lower processing complexity than other unsupervised methodologies [23]. Some unsupervised methodologies confine their analysis to the magnitude component of the SCVs, leading to a 1-D problem and a considerable reduction in information regarding the character of the change [26], [27]. Another approach involves the analysis of SCVs in which only two spectral channels are chosen [24]. However, it is important to note that this approach still demands a prior understanding to select the optimal two-band combination that encompasses essential details about the particular change in question [22]. Another study proposed an unsupervised framework for the conversion of SCVs into a feature space known as Compressed CVA (C^2VA). This transformation incorporated all spectral channels, rendering it unnecessary to possess prior knowledge regarding the change type [12]. To the best of our knowledge, the existing literature has primarily employed CVA and C^2VA in the context of bitemporal and multitemporal images for CD [12], [24], [28], [29]. This application necessitates the careful selection of preimage and postimage to highlight the change, which, in turn, requires knowledge of the timing of the change. In this context, there is still significant potential for research advancement to exploit the abundant change-related data embedded in the SCVs and to extend the application of these approaches to CD in SITS.

SITS can become denser in both temporal and spectral dimensions (i.e., high temporal and spectral resolution). Utilizing SITS data for CD offers numerous advantages: 1) it diminishes the need to perform a supervised selection of a couple of images to illustrate the change; 2) by incorporating a greater quantity of images spanning multiple time frames (increasing temporal density), it offers a more extensive dataset, thereby aiding in the identification of diverse change events occurring at various time intervals; 3) from the spectral perspective, SITS allows for the detection of recurrent changes with distinct spectral characteristics; 4) SITS offers the potential to detect seasonal, gradual, and abrupt changes, thus enhancing the comprehension of land cover dynamics from a broader perspective [16]; and 5) SITS facilitates practical and continuous monitoring of land cover changes. Despite these advantages, CD in SITS presents a greater challenge than the bitemporal context, primarily due to the inherent complexity resulting from the higher dimensionality of the time series data. In addition, the information embedded within time series is susceptible to interference from residual geometric inaccuracies, atmospheric scattering, and cloud-related effects [16]. Various methodologies have been explored in the literature to deal with CD in SITS [14], [16], [21], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39]. However, utilizing the full dimensionality of SITS is still challenging, especially when dense SITS is considered [40]. Effective handling of such data requires robust computational capabilities and a sophisticated processing algorithm capable of efficiently managing and extracting the most relevant information/dimensions for CD. An approach that can minimize dimensions, thus lowering the computational resources necessary for processing the many images within SITS, without compromising on the depth of information, is needed.

This article introduces time series CVA (TSCVA), an exploratory approach for pixel-based detection of multiple changes in dense SITS. TSCVA leverages spectral, temporal, and combined spectral–temporal information within the time series, analyzing these components both independently and in conjunction. The method requires some semisupervision in selecting clusters linked to changes through photograph interpretation, but does not require labeled training samples. Building on the foundation laid by CVA [22], which ensures robustness under sparse or irregular acquisitions, and on its extension C^2VA, which introduced an unsupervised clustering framework, our approach redefines C^2VA within the time series feature space. As a result, TSCVA can adapt to different temporal data densities: when observations are limited, it naturally reduces to the CVA case in bitemporal settings, whereas with denser time series, it can exploit the additional temporal information to improve sensitivity to gradual dynamics such as seasonal shifts. Furthermore, it introduces new criteria for characterizing changes in both the magnitude and direction of the time series SCVs. The processing of time series provides us with the means to integrate the abundant spectral–temporal data found in satellite imagery, enabling a more comprehensive exploration of change components. The novel framework for C^2VA adapted to time series analysis makes use of magnitude and direction, which contain valuable insights for the identification of changes in both spectral characteristics and temporal patterns. The spectral dimension

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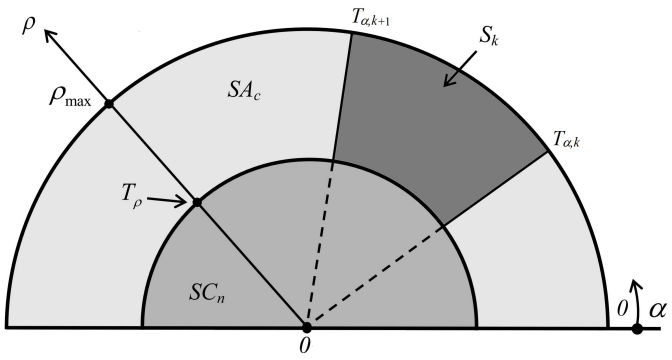


Fig. 1. 2-D polar representation of the multi-CD problem of C^{2}VA [12].

aids in detecting changes characterized by specific features that may be correlated with one or a few spectral channels, while the temporal dimension provides valuable clues for recognizing particular changes based on their timing. The objective of this article is to present an innovative pixel-based framework for C^{2}VA in time series analysis and to offer an initial demonstration of its effectiveness in capturing spectral–temporal characteristics of changes. Our approach emphasizes leveraging the distinct information embedded in the spectral and temporal components, both individually and in combination.

The rest of this article is organized as follows. Section II reviews the C^{2}VA state-of-the-art method in bitemporal images. The proposed semisupervised multiple-CD (multi-CD) framework (i.e., TSCVA) is described in Section III. Section IV introduces the SITS data utilized in this research and the experimental settings, followed by an in-depth analysis of the experimental results. Finally, in Section V, we summarize and conclude the findings presented in this work.

II. Compressed CVA (C^{2}VA) in Bitemporal Images

Let us consider two multispectral images X_1 and X_2 acquired over the same geographical region at two different times t_1 and t_2, respectively. Let D be the multispectral difference image computed by pixelwise radiometric subtraction of X_1 from X_2

$$D = X_2 - X_1. \tag{1}$$

Let $\Omega = \{\omega_n, \Omega_c\}$ be the set of classes to be identified in D. ω_n represents the class of no-change pixels, and $\Omega_c = \{\omega_{c1}, \omega_{c2}, \ldots, \omega_{cK}\}$ is a meta-class that gathers all the possible kinds of changes K. D represents the SCVs, which can belong to either Ω_c or ω_n. The objective of the multi-CD is to identify the changed pixels (Ω_c) and to distinguish between multiple change classes $\{\omega_{c1}, \omega_{c2}, \ldots, \omega_{cK}\}$.

In C^{2}VA, the SCVs can be represented by two features for CD purposes, namely the magnitude ρ and the direction α [12]

$$\rho = \sqrt{\sum_{b=1}^{B} (D_b)^2} = \sqrt{\sum_{b=1}^{B} \left(X_{b,2} - X_{b,1}\right)^2} \tag{2}$$

$$\alpha = \arccos\left(\sum_{b=1}^{B} (D_b R_b) \Big/ \sqrt{\sum_{b=1}^{B} (D_b)^2 \sum_{b=1}^{B} (R_b)^2}\right) \tag{3}$$

where D_b and R_b are the bth ($b = 1, \ldots, B$) band of D and the reference vector R, respectively. B represents the number of spectral channels in $X_{b,1}$ and $X_{b,2}$. ρ provides insight into whether changes are evident or not. Pixels representing changes tend to have higher values, whereas those representing a lack of change tend to have lower values [12], [24]. The process of determining the direction α entails assessing the angular separation between each SCV in the D space and the reference vector R. This procedure yields valuable information for distinguishing between various change classes $\{\omega_{c1}, \omega_{c2}, \ldots, \omega_{cK}\}$. In the context of the considered image pair, the two variables, ρ and α, define a 2-D polar coordinate system where $\rho \in [0, \rho_{\max}]$ and $\alpha \in [0, \pi]$, with $\rho_{\max}$ and π being the maximum values for ρ and α, respectively. A 2-D domain with a semicircular form can be used to illustrate all the SCVs within the D space, as shown in Fig. 1.

In the C^{2}VA framework, the vector R is defined as a constant unit vector with a length of B, represented by $R = [1/\sqrt{B}, \ldots, 1/\sqrt{B}]$. This setup ensures equal weighting of all spectral channels in the CD process. However, a notable drawback of this approach is the potential introduction of ambiguity due to information compression, which arises from the simplified representation using α. This compression can result in similar α values across different change categories, thereby increasing uncertainty [12].

The state-of-the-art C^{2}VA [12] addresses the challenge of the multi-CD problem through a two-step process, both of which involve thresholding.

1) With respect to ρ, a threshold value T_ρ, the semicircle separates into two parts, known as SC_n, which corresponds to the semicircle representing ω_n, and SA_c, which corresponds to the semiannulus representing Ω_c, as shown in Fig. 1

$$\mathrm{SC}_n = \left\{\rho, \alpha | 0 \le \rho < T_\rho \quad \text{and} \quad 0 \le \alpha \le \pi\right\} \tag{4}$$

$$\mathrm{SA}_c = \left\{\rho, \alpha | T_\rho \le \rho \le \rho_{\max} \quad \text{and} \quad 0 \le \alpha \le \pi\right\}. \tag{5}$$

2) Through an examination of the semiannular region within SA_c along the α direction, it becomes possible to differentiate between $\{\omega_{c1}, \omega_{c2}, \ldots, \omega_{cK}\}$ by utilizing $K-1$ angular thresholds $T_{\alpha,k} (k = 1, \ldots, K-1)$

$$S_k = \left\{\rho, \alpha | T_\rho \le \rho \le \rho_{\max} \quad \text{and} \quad T_{\alpha,k} \le \alpha \le T_{\alpha,k+1}\right\} \tag{6}$$

where $0 \le T_{\alpha k} < T_{\alpha,k+1} \le \pi$.

C^{2}VA has showcased its effectiveness in handling the problem of multi-CD within bitemporal images [2], [12]. The use of bitemporal images for CD requires a supervised process that entails the adequate selection of a pair of images that most accurately captures the change phenomenon. This, in turn, necessitates a clear understanding of when the change took place. Furthermore, this method confines the CD process to identifying changes occurring between the two chosen images only, and thus to abrupt changes.

III. Time Series CVA

With time series analysis, supervised image selection is no longer necessary, as it makes use of a larger collection of images. This expanded dataset enhances the capacity to detect a wider range of changes, including both intraannual and

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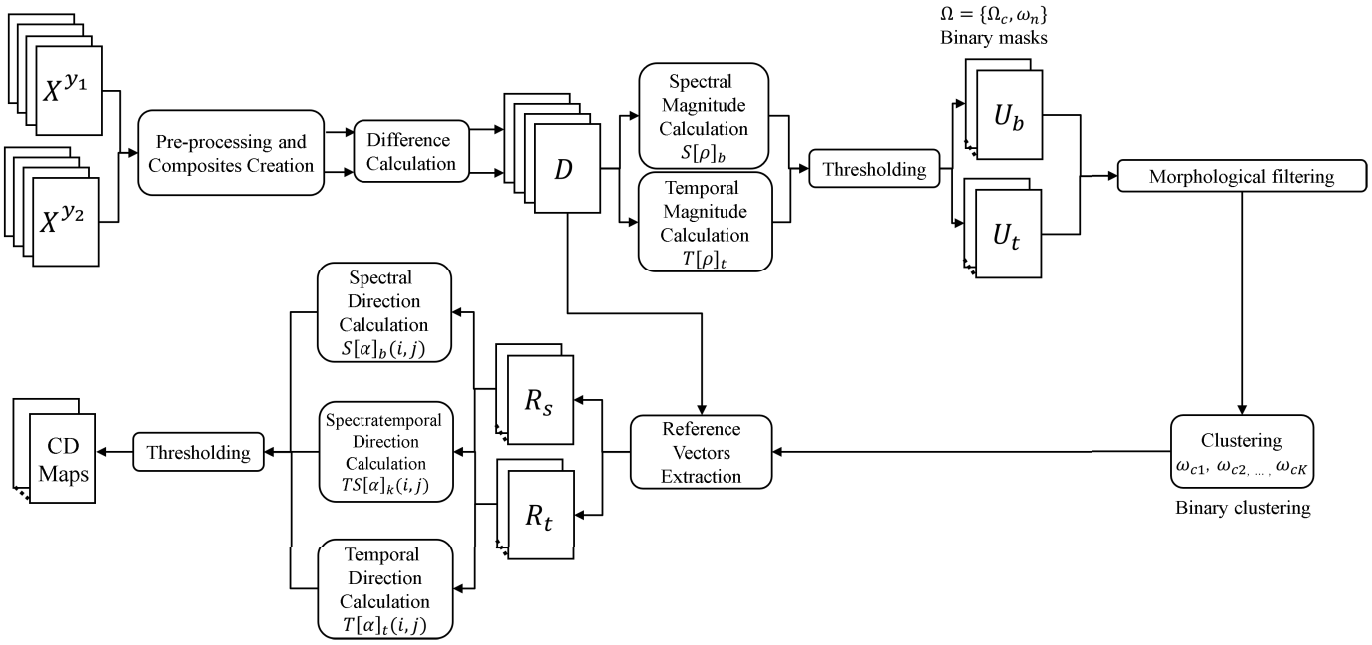


Fig. 2. Block scheme of the proposed TSCVA.

gradual inter-annual changes, as well as sudden changes [14], [16]. In SITS, the change classes can be classified into two categories: spectral and temporal changes. Spectral changes refer to changes in the spectral reflectance, indicating transformations in land cover types. Temporal changes involve the evolution of land cover changes over time, highlighting shifts and patterns as they develop across different time intervals.

To utilize the extensive change information available in SITS and to deal with the aforementioned problems when applying C^{2}VA for CD in SITS, we theoretically redefine C^{2}VA in the time series feature space. TSCVA centers on minimizing the ambiguity associated with C^{2}VA by crafting a unique reference vector, thereby implementing zero compression in the direction variable. In addition, TSCVA is not constrained by a fixed monitoring frequency or a uniform number of acquisitions per year. The method can therefore be applied to irregular or incomplete time series, where the number of available observations may vary across years, while naturally reducing to the CVA formulation in the bitemporal case. This flexibility ensures that TSCVA remains applicable in real-world settings, where cloud cover, data gaps, or irregular acquisition schedules often limit the availability of dense time series. The block scheme of the proposed CD framework is shown in Fig. 2. Let $X^y_{t,b}$ be the image representing the tth ($t = 1, \ldots, T$) and bth ($b = 1, \ldots, B$) component of the time series S^y, where the superscript y indicates the acquisition year, t denotes the acquisition date within the year, and b denotes the number of spectral channels

$$S^y_{t,b} = \begin{bmatrix} X^y_{1,1} & \cdots & X^y_{1,B} \\ \vdots & \ddots & \vdots \\ X^y_{T,1} & \cdots & X^y_{T,B} \end{bmatrix}. \tag{7}$$

Let us consider two co-registered time series S^{y_1} and S^{y_2}, acquired by the same sensor over the same geographical region in two different years: y_1 and y_2, respectively. TSCVA considers a pixelwise radiometric comparison between S^{y_1} and S^{y_2}, inspired by C^{2}VA [12], but expanded to accommodate SITS with more than two images. The radiometric evaluation involves subtracting the spectral feature vectors linked with each corresponding spatial location across the two time series images S^{y_1} and S^{y_2}

$$S^{y_2}_{t,b} - S^{y_1}_{t,b} = D_{t,b} \tag{8}$$

$$D_{t,b}(i,j) = \begin{bmatrix} D_{1,1}(i,j) & \cdots & D_{1,B}(i,j) \\ \vdots & \ddots & \vdots \\ D_{T,1}(i,j) & \cdots & D_{T,B}(i,j) \end{bmatrix}. \tag{9}$$

Each pixel in $D_{t,b}$ can be illustrated as an SCV with $t \cdot b$ components, where $D_{t,b}(i,j)$ is the pixel component at time t and spectral band b. Let us define the time series magnitude as TS[ρ] and the time series direction as TS[α]. In TSCVA, the objective is to investigate these two components to identify significant changes by understanding the evolution of the SCVs across the temporal dimension. The resulting time series difference $D_{t,b}$ comprises the same number of images within S^{y_1} and S^{y_2}. To understand each change component thoroughly and to detect changes, we analyze TS[ρ] and TS[α] from two frames of reference: spectral and temporal. Analyzing TS[ρ] and TS[α] provides invaluable clues for detecting different types of changes.

The simplified representation of the fixed unit reference vector $R = [1/\sqrt{B}, \ldots, 1/\sqrt{B}]$ used to calculate α in C^{2}VA does not allow for the discrimination of the changes available within the time series, leading to a substantial loss of information. To address this challenge, an alternative selection of R needs to be established. This choice of R possesses the capability to autonomously optimize the differentiation of the change classes in an unsupervised manner. To accomplish this

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objective, TSCVA incorporates three analytical steps: 1) binary CD based on the time series magnitudes to distinguish between ω_n and Ω_c; 2) selection of reference matrices; and 3) characterization of Ω_c based on the time series directions.

A. Binary CD Based on Time Series Magnitudes

The magnitude quantifies the degree of change [3], [26]. The temporal magnitude $T[\rho]_t$ is computed by fixing the temporal component t (i.e., there are $TT[\rho]_t$). The squared difference between each change vector is calculated for every spectral band and then summed up over all spectral channels B of the SITS before being square rooted

$$T\left[\rho\right]_t = \sqrt{\sum_{b=1}^{B}\left(D_{t,b}\right)^2}. \tag{10}$$

Fixing the temporal component t focuses on the change information contained in the temporal domain. $T[\rho]_t$ represents specific change information correlated with their time of occurrence.

Similarly, the spectral magnitude $S[\rho]_b$ is computed by fixing the spectral component b (i.e., there are $BS[\rho]_b$). The squared difference between each change vector is calculated for every time acquisition and then summed up over all temporal time frames T of the SITS before being square rooted

$$S\left[\rho\right]_b = \sqrt{\sum_{t=1}^{T}\left(D_{t,b}\right)^2}. \tag{11}$$

Fixing the spectral component b focuses on the change information contained in the spectral domain. $S[\rho]_b$ represents specific change information correlated with one or a few spectral channels, making it extra visible when observed from the spectral element point of view, and changes that do not span over multiple time frames.

The magnitudes are consistently represented as 2-D images. If the SCV $T[\rho]_t$ or $S[\rho]_b$ is diminutive or nearing zero, the associated pixel is classified into the no change class. Conversely, when the $T[\rho]_t$ or $S[\rho]_b$ of the SCV is substantial, the pixel is categorized as belonging to the change class [24]. Utilization of automated thresholding strategies to classify pixels into the two classes [i.e., change (Ω_c) and no change (ω_n)] is preferred in the literature [3], [26], [27], [41]. When considering spatial independence and the normal distribution of pixels, it is plausible to represent the distribution of magnitudes with the Rayleigh-Rice mixture density [42]. The probability distribution of ω_n is described by the Rayleigh distribution, while that of Ω_c is characterized by the Rice distribution. The statistical distributions for the Ω_c and ω_n classes are estimated using the expectation–maximization (EM) algorithm, which determines the parameters of the Rayleigh-Rice mixture density, as evidenced in prior literature [12], [42]. In addition, the Bayes decision theory is implemented to compute the thresholds $T_T[\rho]_t$ and $T_S[\rho]_b$ for $T[\rho]_t$ and $S[\rho]_b$, respectively, to segregate the two classes. Thresholding is employed to demonstrate the framework in a controlled setting; however, alternative decision-making methods and other thresholding techniques from existing literature, such as those discussed by Bruzzone and Prieto [26], [27], are applicable. Applying the threshold, preliminary binary CD maps M_x are obtained where x denotes the fixed frame of reference [i.e., time (t) and band (b)] and (i, j) identifies the spatial position of the pixels within each magnitude image. M_x represents the initial segmentation or detection results based on the specified threshold values

$$M_x(i,j) = \begin{cases} 1, & \text{if } \left[\rho\right]_x(i,j) > T_{\left[\rho\right]_x} \\ 0, & \text{otherwise.} \end{cases} \tag{12}$$

B. Extraction of Reference Matrices

In contrast to C^2VA, which utilizes a fixed reference vector $R = [1/\sqrt{B}, \ldots, 1/\sqrt{B}]$, TSCVA proposes the use of a change-dependent reference vector R_k for $k = 1, \ldots, K$, where K is the number of changes. The selection of R_k is based on a unique value clustering method. The preliminary CD maps M_x, obtained from the thresholding process, are then subjected to a series of morphological filtering operations. The refined maps, still denoted by M_x*, are subsequently used in constructing the final maps U_x. Let U_x be the union of binary CD maps obtained by stacking the binary CD maps M_x*

$$U_x = [M_1{*}, \ldots, M_x{*}] \tag{13}$$

where $x = t, b$.

The set U_x consists of binary vectors. We apply morphological filtering to each M_x* in U_x to generate reference matrices to measure angular distances. This filtering process includes area filtering to remove isolated pixels, which are considered noise, and erosion, which thins the binary images using a specific structuring element. The filter size is chosen based on the expected size and shape of the changes and the desired balance of false alarms and missed detections [43], [44]. Erosion operation systematically reduces small, noisy regions while preserving the overall structure of changes. The process ensures that statistically significant contiguous change regions remain while eliminating isolated pixel-level detections that may arise due to artifacts or residual misalignment. In addition, we refine the CD maps using a connectivity-based sieving procedure designed to remove small, isolated detections. Each pixel is retained only if a sufficient number of neighboring pixels (with neighborhood size adjustable according to analysis requirements) are also classified as changed. The larger the selected structural element or neighborhood criterion, the greater the number of isolated pixels excluded. This approach effectively reduces false positives (FPs) caused by noise or spectral artifacts while preserving genuine areas of change. The morphological filtering operates on the intermediate binary CD maps M_x prior to the unique value clustering step, with the purpose of suppressing isolated noisy pixels and stabilizing the clustering process. This pre-processing does not smooth the TSCVA outputs, which are derived from the spectral–temporal integration of the time series and thus preserves the spatial detail information. The improved performance of TSCVA is therefore primarily attributable to its ability to exploit both spectral and temporal dimensions, with morphological filtering serving only as a noise-reduction step. After morphological treatment, the refined binary CD maps

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are stacked sequentially to form U_x. To determine R_k unsupervised, the clustering is applied to U_x. This method of selecting the reference vector involves distinguishing between different binary vectors to classify the types of changes within U_x. Each cluster k corresponds to a change (i.e., $\omega_{c1}, \omega_{c2}, \ldots, \omega_{cK}$). Given that U_t is composed of binary CD maps generated at a constant t value, resulting in T layers, the clusters formed in this situation emphasize the temporal aspects of the changes, with a particular focus on when changes occur. In contrast, U_b comprises binary CD maps generated when the spectral component is fixed, resulting in B layers. The isolation of unique pixel values across them allows for a concentration of the spectrally evident changes. R_k is computed as the average pixel values of changed pixels Ω_c within the difference time series $D_{t,b}$ associated with each cluster $k(k = 1, \ldots, K)$

$$R_k = \frac{1}{D_{\Omega_c}} \sum_{(i,j)|M_x(i,j)=1}^{D_{\Omega_c}} D_{t,b}(i,j). \tag{14}$$

C. Time Series Direction Analysis

Unlike C^{2}VA, which compresses information from all spectral bands into two dimensions and uses a single reference vector to calculate angular distances for each change vector, TSCVA employs a different method. It computes reference matrices, which project the time series direction into an optimal 2-D space for each spectral and temporal component, allowing for a more precise determination of change types. The selection of R_k enables a customized computation of angular distances, as R_k is derived from the average pixel values of pixels that have undergone changes within $D_{t,b}$ for each $k(k = 1, \ldots, K)$. Each pixel (i, j) in $D_{t,b}$ is assigned to the cluster k^* that minimizes the angular distance. The pixels belonging to the kth cluster exhibit smaller angular distances, while all other pixels show larger values. R_k is determined semisupervised, guided by visual analysis and prior knowledge of observed changes.

The direction analysis $\mathrm{TS}[\alpha]_k$ is conducted to model the behaviors associated with various kinds of changes, where k corresponds to a particular change (i.e., $\omega_{c1}, \omega_{c2}, \ldots, \omega_{cK}$). Consequently, it can be deduced that varying alpha values will characterize distinct types of changes. The computed angle signifies the degree of resemblance between the change vector $D_{t,b}$ and the reference vector R_k. Computing $\mathrm{TS}[\alpha]_k$ also reduces noise associated with the kth change class, resulting in a lower rate of false alarms. $\mathrm{TS}[\alpha]_k$ extends the direction calculation employed in C^{2}VA [12] to align with the time series context, using a change-dependent reference vector while avoiding compression. The computation of this direction is carried out separately in the spectral, temporal, and spectra-temporal domains. The time series temporal direction $T[\alpha]_t$ can be computed by fixing the time component t (resulting in $TT[\alpha]_t$)

$$T[\alpha]_t = \arccos\left(\sum_{b=1}^{B}\left(D_{t,b}R_k\right) \Big/ \sqrt{\sum_{b=1}^{B}\left(D_{t,b}\right)^2 \sum_{b=1}^{B}(R_k)^2}\right). \tag{15}$$

Conversely, the time series spectral direction $S[\alpha]_b$ is determined by fixing the spectral component b (leading to $BS[\alpha]_b$)

$$S[\alpha]_b = \arccos\left(\sum_{t=1}^{T}\left(D_{t,b}R_k\right) \Big/ \sqrt{\sum_{t=1}^{T}\left(D_{t,b}\right)^2 \sum_{t=1}^{T}(R_k)^2}\right). \tag{16}$$

Finally, the spectral–temporal direction $\mathrm{TS}[\alpha]_k$ is obtained by considering the entire multispectral difference images $D_{t,b}$ (involving all the spectral channels B and the temporal time frames T), as follows:

$$\mathrm{TS}[\alpha]_k = \arccos\left(\sum_{t=1}^{T}\sum_{b=1}^{B}\left(D_{t,b}R_k\right) \Big/ \sqrt{\sum_{t=1}^{T}\sum_{b=1}^{B}\left(D_{t,b}\right)^2 \sum_{t=1}^{T}\sum_{b=1}^{B}(R_k)^2}\right). \tag{17}$$

IV. Experimental Evaluation

We validated TSCVA on two Sentinel-2 Level 2A surface reflectance time series datasets comprising three major changes: wildfires, deforestation, and changes in lake surface. The first dataset corresponds to the Kineta wildfire in Greece (tile 34SFH), consisting of 86 cloud-screened scenes from January to December 2017 and 117 from January to December 2018. The second dataset focuses on deforestation in California's Stanislaus National Forest (tile 10SGH), comprising 32 images from May to November 2018 and 33 from May to November 2019, reflecting the reduced availability of usable acquisitions outside the snow-free season. Both datasets are characterized by a spatial resolution of 10 m for $b2$, $b3$, $b4$, and $b8$, and 20 m for $b5$, $b6$, $b7$, $b8a$, $b11$, and $b12$, where b corresponds to the bands of Sentinel-2. For comparison purposes, we applied BFAST to the same time-series lengths and performed standard CVA on two image pairs per dataset: 1) The first pair comprises two images chosen at random, mimicking scenarios where there is no prior knowledge about the timing of the change and 2) The second image pair is selected based on prior knowledge regarding the timing of the detected change (a priori information). This is the optimal working condition, and the results can be considered as a performance upper bound. Since TSCVA is inherently pixel-based, comparisons were conducted against the pixel-based implementations of BFAST and CVA to ensure methodological consistency. The evaluation of the effectiveness of TSCVA was conducted based on the following criteria: 1) a qualitative assessment involved comparing the abilities of TSCVA, BFAST, and CVA to represent change information or 2) a quantitative analysis conducted to measure the performance of the proposed technique in multi-CD (based on a reference map) applied to BFAST, CVA, and TSCVA. Performance metrics include FP, false negative (FN), omission errors (OEs), and the $F1$ score. We show that TSCVA overcomes the need for a supervised image selection and can perform multi-CD

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(a) (b) (c)

Fig. 3. True color composites (RGB) of the Greece dataset acquired by Sentinel-2 in (a) April–June 2017, (b) April–June 2018, and (c) reference map (black: no change, white: change in t3, and gray: masked-out water).

semisupervised, exploiting a larger number of images from the time series.

The Sentinel-2 Level 2A product (surface reflectance) was subjected to spatial down-sampling to 20 m resolution, radiometric correction, and masking of clouds, snow, and sea water (specific to the Greece dataset), followed by image co-registration. Directly comparing each pair of images from each time series can be resource-intensive. It may result in an elevated number of false alarms, primarily due to the absence of consideration for seasonal variations, temporal correlations between images, and the loss of pixels caused by filter masks. Given that we have two years of images, S^{y_1} and S^{y_2}, acquired by the same sensor over the same geographical region in two different years, y_1 and y_2, the experimental configuration includes the generation of four seasonal composites per year. For the Greece dataset, the composites cover the full January–December period, whereas for the California dataset, they are restricted to the snow-free months between May and November. Each composite is calculated as the per-pixel median of all valid images within a fixed seasonal interval, thereby preserving the temporal structure of the time series

$$S^{y_1} = \left[X^1_{1,b}, X^1_{2,b}, X^1_{3,b}, X^1_{4,b}\right]$$
$$S^{y_2} = \left[X^2_{1,b}, X^2_{2,b}, X^2_{3,b}, X^2_{4,b}\right]. \quad (18)$$

In this context, $t(t = 1, 2, 3, 4)$ denotes the seasonal composite, and b ($b = b2$, $b3$, $b4$, $b5$, $b6$, $b7$, $b8$, $b8a$, $b11$, and $b12$) refers to the Sentinel-2 bands. This is the result of cloud and quality filtering and of seasonal composites calculation obtained from the available images of each year. Time series in (18) are the input to the CD process.

A. Greece Dataset

1) Experimental Setup: The first dataset comprises two years of images acquired from the Kineta wildfire in the west of Athens, Greece, covering the full period from January to December in both 2017 and 2018. The data were collected by the Sentinel-2 satellite. The selected study area is a cropped section of tile 34SFH (1076 × 1560 pixels). A wildfire heavily affected the Mediterranean shrubland, reducing it to burned undergrowth on July 23, 2018. Fig. 3 illustrates the true color composites (b4, b3, and b2) for April–June 2017 [Fig. 3(a)], 2018 [Fig. 3(b)], and a reference map [Fig. 3(c)], obtained

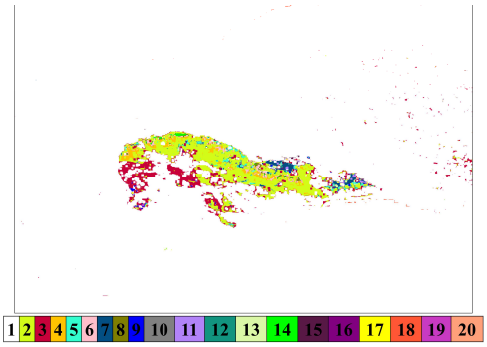


Fig. 4. Union CD map U_b of the Greece dataset consisting of 20 unique value clusters.

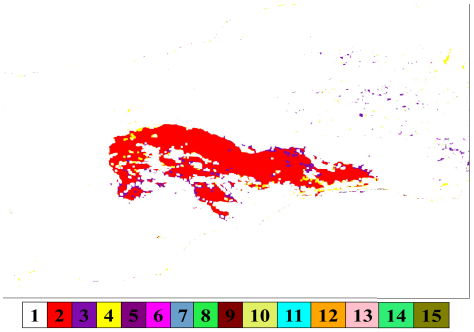


Fig. 5. Union CD map U_t of the Greece dataset consisting of 15 unique value clusters.

by a comprehensive visual analysis of the time series images and incorporation of available prior knowledge related to the change. The acquired reference map contains 154 680 changed pixels and 1523 880 unchanged pixels. The delineation was performed by visually interpreting spectral changes in shortwave infrared (SWIR) and NIR bands and cross-validating against the documented wildfire event date, ensuring temporal consistency across the time series.

2) Results: By implementing the proposed framework in both spectral and temporal domains, we identified 109 distinct spectral clusters and 15 unique temporal clusters. These clusters captured changes related to the wildfire and its varying severity levels, as well as other regional changes. Fig. 4 presents the union CD map U_b, which includes 20 spectral

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TABLE I

NUMBER OF PIXEL COUNTS CORRESPONDING TO EACH UNIQUE VALUE IN THE SPECTRAL CLUSTER OF THE GREECE DATASET

Cluster	Pixel value										Pixel count
	b_2	b_3	b_4	b_5	b_6	b_7	b_8	b_{8A}	b_{11}	b_{12}	
R_{s1}	0	0	0	0	0	0	0	0	0	0	1067802
R_{s2}	0	0	0	0	0	0	0	0	1	1	45306
R_{s3}	0	0	0	0	0	0	0	0	0	1	23453
R_{s4}	0	0	1	0	0	0	0	0	1	1	6253
R_{s5}	0	0	0	0	0	1	0	1	1	1	3099
R_{s6}	1	0	1	0	0	0	0	0	1	1	2751
R_{s7}	0	0	0	0	1	1	1	1	1	1	2748
R_{s8}	1	1	1	0	0	0	0	0	1	1	1164
R_{s9}	0	0	0	0	0	1	0	1	0	1	1007
R_{s10}	0	0	0	0	0	0	0	1	1	1	940
R_{s11}	0	0	0	0	1	1	1	1	0	1	869
R_{s12}	0	0	0	0	0	1	0	0	1	1	849
R_{s13}	0	0	0	0	1	1	0	1	1	1	803
R_{s14}	1	1	1	1	0	0	0	0	1	1	703
R_{s15}	0	0	0	0	0	1	0	1	0	0	428
R_{s16}	0	0	0	0	0	1	1	1	1	1	411
R_{s17}	0	0	0	0	1	1	0	1	0	1	375
R_{s18}	1	1	1	1	1	1	1	1	1	1	244
R_{s19}	0	0	0	0	0	1	0	0	0	1	244
R_{s20}	0	0	0	0	0	1	0	0	0	0	244

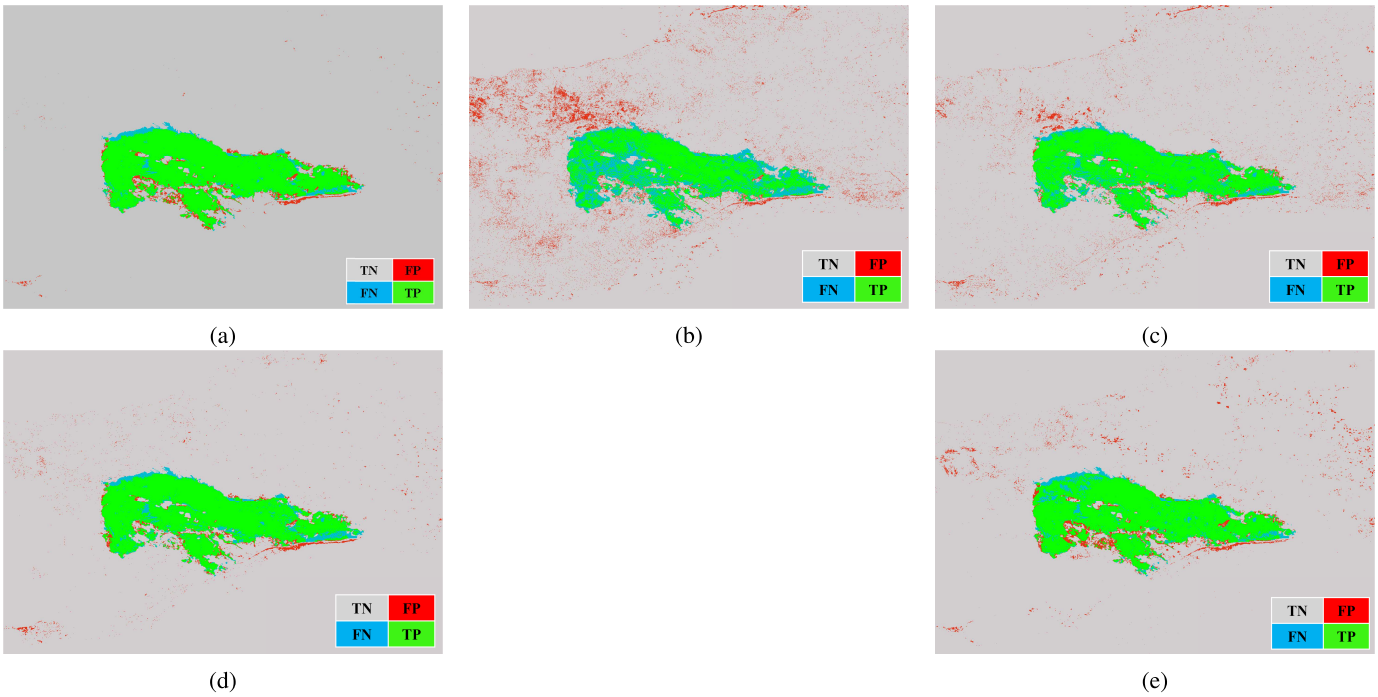


Fig. 6. CD maps of (a) spectra-temporal analysis, spectral analysis of cluster R_{s2} projected at (b) $b2$, (c) $b4$, (d) $b12$, and (e) temporal cluster of R_{t2} projected at $t = 3$ (Greece dataset).

clusters, while Fig. 5 shows the union CD map U_t, containing 15 temporal clusters. Table I details the 20 spectral clusters depicted in Fig. 4, and Table II provides the 15 temporal clusters from Fig. 5, including the pixel counts for each bandwise unique value cluster.

To demonstrate the effectiveness of the framework, we performed a semisupervised selection of representative clusters from each domain using a priori knowledge, enabling a clear illustration of the spectro-temporal analysis. We chose three spectral clusters (R_{s2}, R_{s3}, and R_{s15} from U_b) and four temporal clusters (R_{t2}, R_{t3}, R_{t4}, and R_{t5} from U_t) to illustrate how effectively the corresponding reference vectors can model Mediterranean shrubland to burned undergrowth changes in the spectro-temporal component. The effectiveness of TSCVA in delineating wildfire-impacted areas is evident from Fig. 6, where changed areas are well delineated. Fig. 6(a) provides a qualitative CD map from the spectral–temporal analysis of spectral cluster R_{s2}. This map incorporates both spectral and temporal components, highlighting how changes evolve over time and across spectral bands. Spectral cluster R_{s2} consists of binary CD maps with changed pixels in layers M_{11} and M_{12}, which correspond to the SWIR bands

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TABLE II

NUMBER OF PIXEL COUNTS CORRESPONDING TO EACH UNIQUE VALUE IN THE TEMPORAL CLUSTER OF THE GREECE DATASET

Cluster	Pixel value				Pixel count
	t_1	t_2	t_3	t_4	
R_{t1}	0	0	0	0	1040580
R_{t2}	0	0	1	1	98151
R_{t3}	0	0	1	0	12366
R_{t4}	0	0	0	1	7215
R_{t5}	1	0	0	1	203
R_{t6}	0	1	0	0	118
R_{t7}	1	0	0	0	96
R_{t8}	0	1	1	0	56
R_{t9}	0	1	0	1	44
R_{t10}	1	1	0	1	26
R_{t11}	1	1	1	1	10
R_{t12}	0	1	1	1	9
R_{t13}	1	0	1	1	7
R_{t14}	1	0	1	0	4
R_{t15}	1	1	0	0	4

known for their sensitivity in effectively distinguishing burned areas.

For the spectral analysis: R_{s1} is categorized within the ω_n class (i.e., 0 for all layers of U_b) and thus omitted from further analysis. Fig. 4 illustrates that the wildfire-affected area primarily consists of three major spectral clusters: R_{s2}, R_{s3}, and R_{s4}, which correspond to the different severity levels of the wildfire. Spectral cluster R_{s2} encompasses the largest portion of the area, as reflected by its highest pixel count, and was our primary focus. Similar analyses were also applied to clusters R_{s3} and R_{s4}. Spectral cluster R_{s2} contains changed pixels in layers M_{11} and M_{12}, which correspond to the SWIR bands, while cluster R_{s3} shows changes in M_{12}, and R_{s4} exhibits changes across M_4, M_{11}, and M_{12}. These observations are consistent with the known sensitivity of these spectral bands to fire events. By examining the spectral characteristics of each cluster in detail, we gained a better understanding of which spectral bands are most effective in detecting wildfires. Conducting a spectral analysis using TSCVA allows us to concentrate on the change information embedded within the spectral channels and to extract spectrally distinct details, such as varying levels of wildfire severity.

For the temporal analysis: by applying the proposed framework in the temporal domain, we identified 15 distinct clusters from U_t, including the ω_n class. Fig. 5 presents the union CD map U_t, which includes these 15 temporally unique clusters. The pixel counts for each unique value cluster are detailed in Table II. Temporal cluster R_{t1}, consisting entirely of pixels from the ω_n class (i.e., 0 across all layers of U_t), was excluded from further analysis. Visual inspection revealed that cluster R_{t2} corresponds to the wildfire-affected area. By substituting R_k with R_{t2} in 15, we projected the wildfire-associated angle into the most fitting 2-D space. Because of reference data availability, we document results from cluster R_{t2} only. However, the same analysis can be performed on all clusters. Fig. 6(e) shows the CD map obtained from temporal cluster R_{t2} projected onto the third composite.

For both qualitative and quantitative assessments, we compare the TSCVA results with the reference map to evaluate its accuracy and effectiveness. Fig. 6 displays the CD maps produced by the spectral directional analysis of cluster R_{s2}, projected onto band 2 [Fig. 6(b)], band 4 [Fig. 6(c)], and band 12 [Fig. 6(d)]. As depicted in Fig. 6(d), projecting spectral cluster R_{s2} onto band 12 delivers the highest accuracy, with fewer FPs (red) and FNs (blue), compared to projections onto band 4 [Fig. 6(c)] and band 2 [Fig. 6(b)]. Table III provides a quantitative evaluation of TSCVA spectral analysis performance against the reference map.

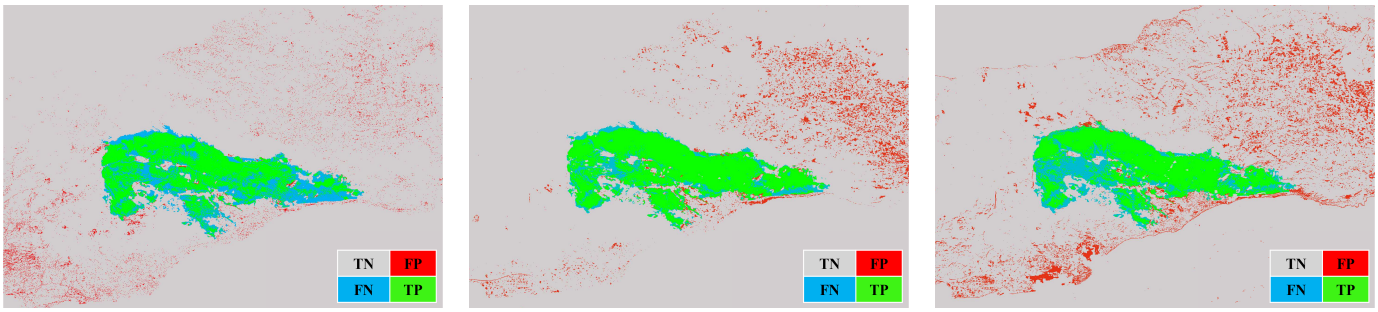
TABLE III

FP, FN, OE, AND $F1$ SCORE METRICS FOR THE SPECTRA-TEMPORAL ANALYSIS, SPECTRALLY FOCUSED TSCVA PROJECTED AT B_2, B_4, AND B_{12}, TEMPORALLY FOCUSED TSCVA, CVA, AND BFAST RESPECTIVELY (GREECE DATASET)

Method	**FP**	**FN**	**OE**	**F1**
Spectra-temporal TSCVA	14966	14112	0.091	0.906
Spectral TSCVA (b2)	56011	55239	0.357	0.641
Spectral TSCVA (b4)	29374	27542	0.178	0.817
Spectral TSCVA (b12)	16200	19587	0.126	0.883
Temporal TSCVA	26679	19363	0.125	0.854
CVA (known timing)	52412	30495	0.199	0.747
CVA (unknown timing)	92232	50155	0.328	0.590
BFAST	28515	59779	0.386	0.682

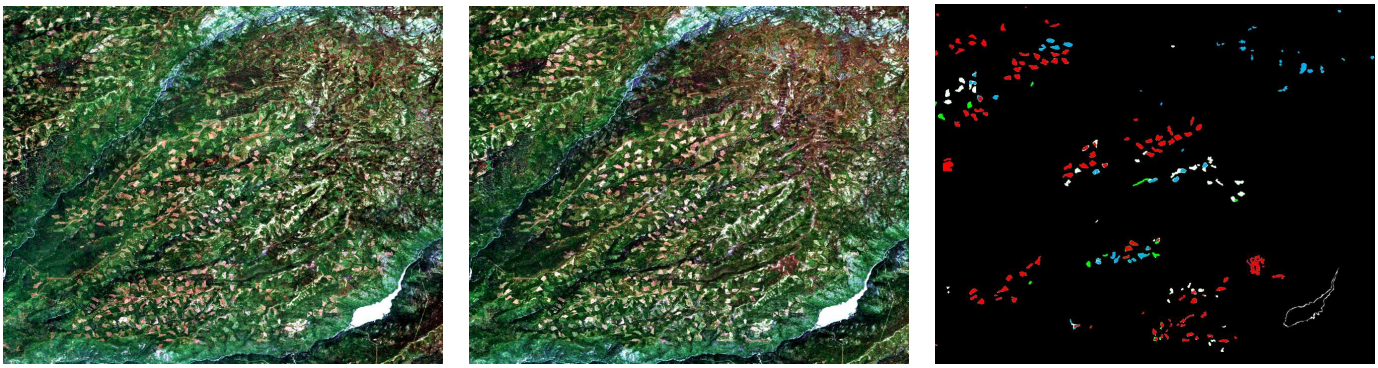
3) Discussion: Through informed reasoning and logical deductions based on available data, we can deduce that 79 out of the 109 spectral clusters are outliers. These clusters exhibit exceptionally low pixel counts compared to others and lack spatial correlations among themselves. The results from the spectro-temporal analysis using spectral cluster R_{s3} show that when only layer M_{12} is changed, the precision in identifying wildfire-affected areas is lower compared to cluster R_{s2}. This aligns with the understanding that changes isolated to a single layer may not capture the whole spectral signature of wildfires in Mediterranean shrubland. In addition, the use of temporal cluster R_{t2}, as outlined in Table II and illustrated in Fig. 5, produces CD maps that distinctly and accurately delineate wildfire-affected regions, confirming the theoretical advantage of combining temporal and spectral information for CD. The effectiveness of spectro-temporal analysis is further illustrated by the improved performance of temporal clusters R_{t4} over R_{t3} in delineating wildfire regions because the wildfire occurred during the third seasonal composite. Some early images in the time series for R_{t3} predate the fire, making R_{t4} more suitable for this specific CD task. TSCVA using temporal cluster R_{t5} provides valuable insights into changes that occurred within the first and fourth composites, as changed pixels correspond to layer M_1 and M_4 of U_t. This rationale applies to all other temporal clusters as well. Our experimental analysis confirms that R_{t2} performs optimally when projected onto the third and fourth composites, as evidenced by the lower number of misclassified pixels, including both FPs and FNs. This timing aligns with the wildfire event on July 23rd, 2018, which falls within the third composite ($X^2_{3,b}$). However, the third composite is slightly less effective than the fourth because the early weeks of July 2018 occurred before the fire, leading to a weaker change signature. Conducting a temporal analysis using TSCVA enables us to concentrate on change information within the temporal domain, allowing for the extraction of

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(a) (b) (c)

Fig. 7. CD maps of (a) BFAST, (b) CVA (known timing), and (c) CVA (unknown timing) (Greece dataset).



(a) (b) (c)

Fig. 8. True color composites of the California dataset acquired by Sentinel-2 in (a) May–June 2018, (b) May–June 2019, and (c) reference map (black: no change, red, blue, white, and green: change in $t1$, $t2$, $t3$, and $t4$, respectively).

details related to the timing of events. By analyzing each cluster temporally, we can distinguish changes based on when they occurred. For instance, R_{t2} shows changed pixels in both the third and fourth composites, which aligns with the timing of the wildfire. In contrast, R_{t3} displays changes in the third composite, while R_{t4} exhibits changes in the fourth composite.

Quantitative evaluation shows that projecting spectral cluster R_{s2} onto band 12 delivers the highest accuracy, with fewer FPs and FNs compared to bands 2 and 4. This is reflected in $F1$ scores of 0.641, 0.817, and 0.883 for bands 2, 4, and 12, respectively. The proposed approach surpasses CVA ($F1$ = 0.590) in situations where the timing of change is unknown, and attains performance comparable to CVA ($F1$ = 0.747) when a priori information about the change timing is known. In addition, a comparison between TSCVA and BFAST reveals that spectra-temporal TSCVA ($F1$ = 0.906) more effectively isolates wildfire disturbances than BFAST ($F1$ = 0.682), as shown in Table III, thereby enabling more accurate CD. The same table also indicates that temporally focused TSCVA ($F1$ = 0.854) outperforms CVA with unknown dates ($F1$ = 0.590), as reflected by higher $F1$ scores. Qualitative comparisons presented in Figs. 6 and 7 further support these findings. Overall, TSCVA delivers results on par with state-of-the-art methods when change timing is known, but without having any prior information on the change time. These results collectively demonstrate that TSCVA outperforms both BFAST and CVA approaches, both quantitatively and qualitatively.

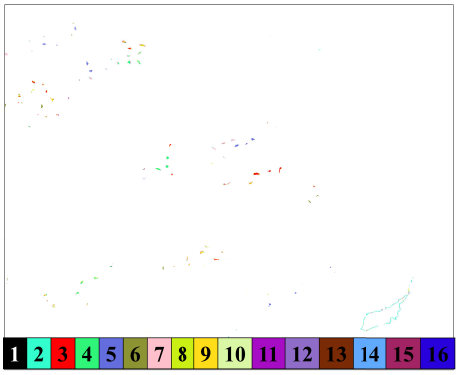


Fig. 9. Union CD map U_t of the California dataset consisting of 16 unique value clusters.

B. California Dataset

1) Experimental Setup: The second dataset comprises biannual time series data obtained from a forest area in the Stanislaus National Forest, California, USA, for the years 2018 and 2019. This data was acquired using the Sentinel-2 satellite. The region under observation is recognized for its tree harvesting activities, primarily for the production of Christmas trees (i.e., a change is observed from evergreen needleleaf forest to bare soil). The area also includes a lake being artificially regulated and thus showing surface fluctuation over time (i.e., a change from water to bare soil is observed). Owing to the substantial snowfall in the area, the two years of images

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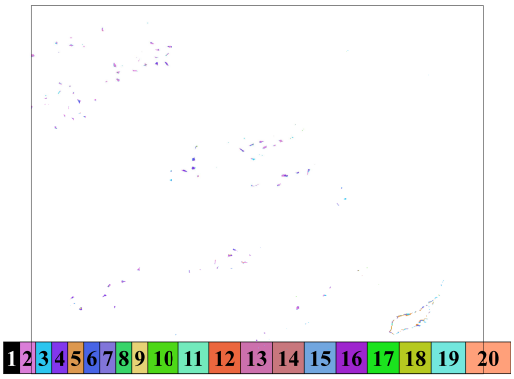


Fig. 10. Union CD map U_b of the California dataset consisting of 20 unique value clusters.

are limited to images captured between May and November for each designated year. The selected study area is a cropped section of the Sentinel-2 tile (10SGH), confined to orbit 70 to further ensure uniformity, with a pixel count of 1515 by 1231. The true color composites for May–June 2018 and 2019 are presented in Fig. 8(a) and (b), respectively. Fig. 8(c) displays a reference map derived from an extensive visual interpretation of the data. This map illustrates the temporal progression and spatial pattern of changes observed throughout the biannual time series. The resulting reference map identifies 33 396 changed pixels and 1831 569 unchanged pixels. The map was produced by interpreting deforestation patches and lake surface changes across seasonal composites, with delineation refined using vegetation red-edge and NIR bands to ensure accurate identification of vegetation stress and water surface variation.

2) Results: Similar to the previous dataset, the spectra-temporal analysis utilized change representative vectors derived from the entire dimensionality of the time series, considering both spectral and temporal domains simultaneously. By applying the proposed framework in both the spectral and temporal domains, we identified 59 unique spectral clusters and 16 unique temporal clusters. These clusters captured changes related to deforestation and its progression over time, lake surface decrement, as well as other regional changes. Figs. 9 and 10 display the union CD maps U_t and U_b, respectively, of the California dataset. Fig. 11(a) qualitatively demonstrates the performance of the spectra-temporal analysis of the spectral cluster R_{s2}. Spectral cluster R_{s2} consists of binary CD maps with changed pixels in layers M_{11} and M_{12}. These bands correspond to the SWIR band. In addition, we performed a spectro-temporal analysis using the change representative matrix obtained from the temporal cluster R_{t2}, where the second composite contains the changed pixel. The resulting CD map illustrates the deforestation patches that occurred within the second composite.

The temporal analysis yields 16 unique temporal cluster values, including the ω_n category, as presented in Table IV. Fig. 9 depicts the union CD map U_t, which integrates the 16 temporally unique clusters. Fig. 8(c) illustrates the progression of deforestation patches throughout the time series. Fig. 9 reveals that the cluster R_{t2} is instrumental in delineating the change in lake surface (bottom right corner of the study area).

TABLE IV
NUMBER OF PIXEL COUNTS CORRESPONDING TO EACH UNIQUE VALUE IN THE TEMPORAL CLUSTER OF THE CALIFORNIA DATASET

Cluster	Pixel value				Pixel count
	t_1	t_2	t_3	t_4	
R_{t1}	0	0	0	0	1854937
R_{t2}	0	1	0	0	630
R_{t3}	0	0	1	1	530
R_{t4}	1	1	1	0	454
R_{t5}	1	0	0	0	432
R_{t6}	0	0	0	1	332
R_{t7}	1	1	0	0	302
R_{t8}	0	0	1	0	287
R_{t9}	0	1	1	1	278
R_{t10}	1	1	1	1	202
R_{t11}	0	1	1	0	105
R_{t12}	0	1	0	1	29
R_{t13}	1	1	0	1	14
R_{t14}	1	0	1	1	11
R_{t15}	1	0	1	0	7
R_{t16}	1	0	0	1	2

The spectral analysis of the time series data for the California dataset identified 59 unique value clusters. Fig. 10 displays the union CD map U_b, which comprises 20 spectrally unique clusters. Table V details these 20 spectral clusters. By visually comparing U_b with the reference map, we could identify the clusters associated with deforestation areas. We focused on clusters R_{s2} and R_{s5}. The analysis of binary CD maps within U_b for these clusters, particularly for bands 6, 7, 8, and 8A, demonstrates that these bands are notably effective in highlighting deforestation areas. Fig. 11 presents the CD maps resulting from the spectral analysis conducted through TSCVA, validated with the reference label data. The quantitative analysis, detailed in Table VI, demonstrates the effectiveness of the proposed method.

3) Discussion: Due to the absence of comprehensive reference data, the analysis is limited to deforestation and lake surface changes. Through analytical evaluation and deduction based on the data, it was determined that 39 of these 59 clusters can be considered outliers due to their significantly low pixel count and lack of spatial correlation. As expected (and confirmed by our experimental analysis), Fig. 11(a) effectively delineates the areas affected by needleleaf tree deforestation. This supports the hypothesis that the spectral cluster R_{s2} is optimal for spectrally detecting deforested areas. Performing TSCVA on SITS provides a wealth of information, allowing us to distinguish and identify both spectral and temporal changes with minimal supervision. A deeper exploration of TSCVA analysis on the spectral and temporal domains will enhance our understanding of the land cover changes affecting the region.

A visual comparison between the unique value clusters and the reference map reveals that deforestation patches comprise multiple unique value clusters, in contrast to the singular cluster structure observed in wildfire events. This suggests that deforestation exhibits a more complex and diverse temporal signature, occurring across multiple time instances. Our study delved into the temporal directionality of each unique value

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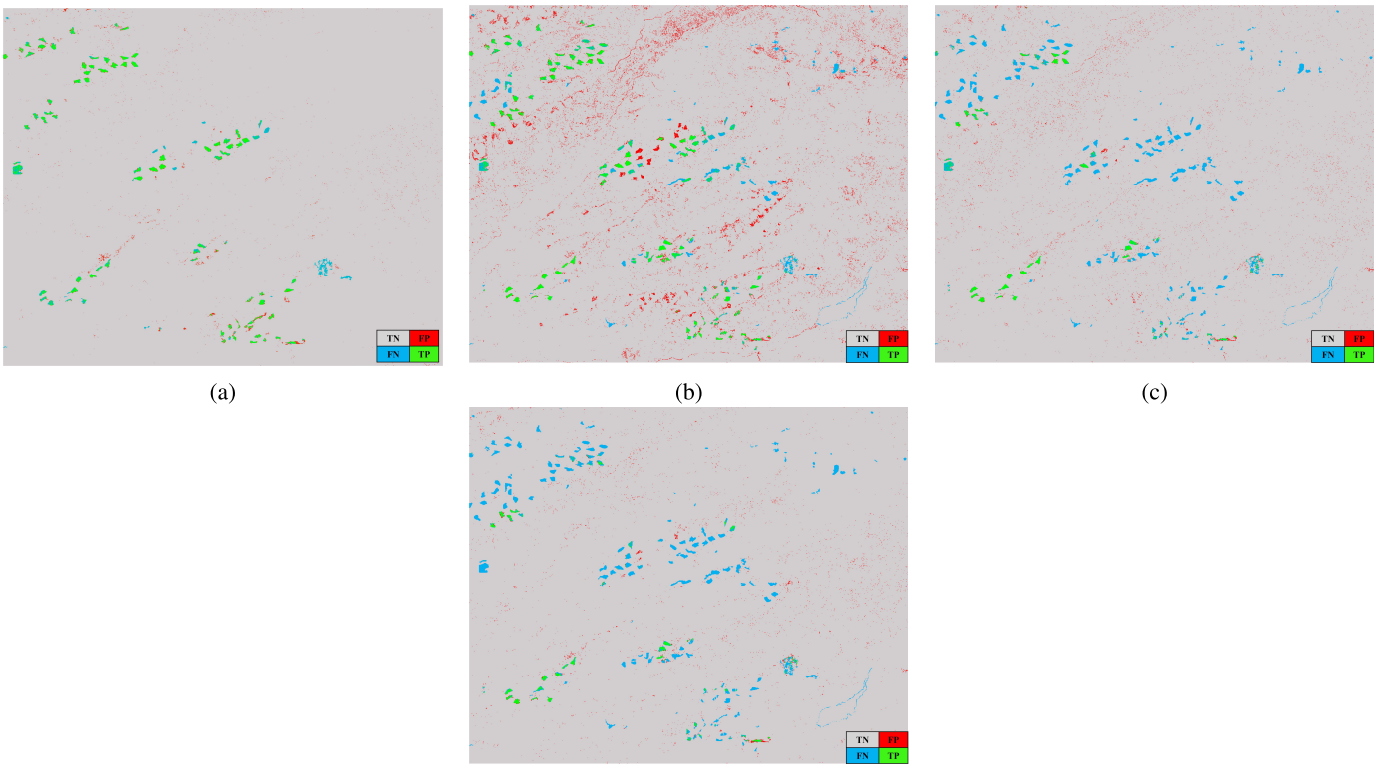


Fig. 11. CD maps of (a) spectra-temporal analysis, and spectral cluster of R_{s2}, projected at (b) $b2$, (c) $b4$, and (d) $b12$, respectively (California dataset).

TABLE V

NUMBER OF PIXEL COUNTS CORRESPONDING TO EACH UNIQUE VALUE IN THE SPECTRAL CLUSTER OF THE CALIFORNIA DATASET

Cluster	Pixel value										Pixel count
	b_2	b_3	b_4	b_5	b_6	b_7	b_8	b_{8A}	b_{11}	b_{12}	
R_{s1}	0	0	0	0	0	0	0	0	0	0	1855084
R_{s2}	0	0	0	0	0	0	0	0	1	1	2022
R_{s3}	0	0	0	0	0	0	0	0	0	1	804
R_{s4}	0	0	1	1	0	0	0	0	1	1	448
R_{s5}	1	1	1	1	1	1	1	1	1	1	414
R_{s6}	0	0	0	1	0	0	0	0	1	1	302
R_{s7}	0	0	0	1	1	1	0	1	1	1	173
R_{s8}	0	0	1	0	0	0	0	0	1	1	100
R_{s9}	0	1	1	1	0	0	0	0	1	1	45
R_{s10}	0	0	0	0	0	0	0	1	0	0	31
R_{s11}	1	1	1	1	1	1	0	1	1	1	31
R_{s12}	0	0	0	0	0	0	0	0	1	0	30
R_{s13}	0	1	1	1	1	1	0	1	1	1	27
R_{s14}	0	0	0	0	0	1	0	1	0	0	18
R_{s15}	0	1	0	1	1	1	0	1	1	1	18
R_{s16}	0	0	0	0	0	0	0	1	0	1	17
R_{s17}	0	0	0	0	0	0	0	1	1	1	14
R_{s18}	0	0	0	1	0	1	0	1	1	1	14
R_{s19}	0	0	0	0	0	1	0	0	0	0	11
R_{s20}	0	0	0	0	1	1	0	1	1	1	11

cluster to examine the temporal dynamics characterizing each deforestation patch and the lake area. The sequential visual representation in Fig. 8(c) clarifies the temporal progression and expansion of deforestation patches and lake surface variation, providing insights into their development over time and indicating the impact of anthropogenic activities. The effectiveness of the vegetation red edge and NIR bands is attributed to their sensitivity to chlorophyll content and vegetation structure, indicating vegetation stress or removal. The spectral analysis through TSCVA allows for a focused examination of spectral changes, enabling the extraction of significant change information based on the specific nature of the spectral data.

To validate our proposed method, we performed a comparative analysis between TSCVA, BFAST comprising the same length of the time series as TSCVA, and conventional CVA

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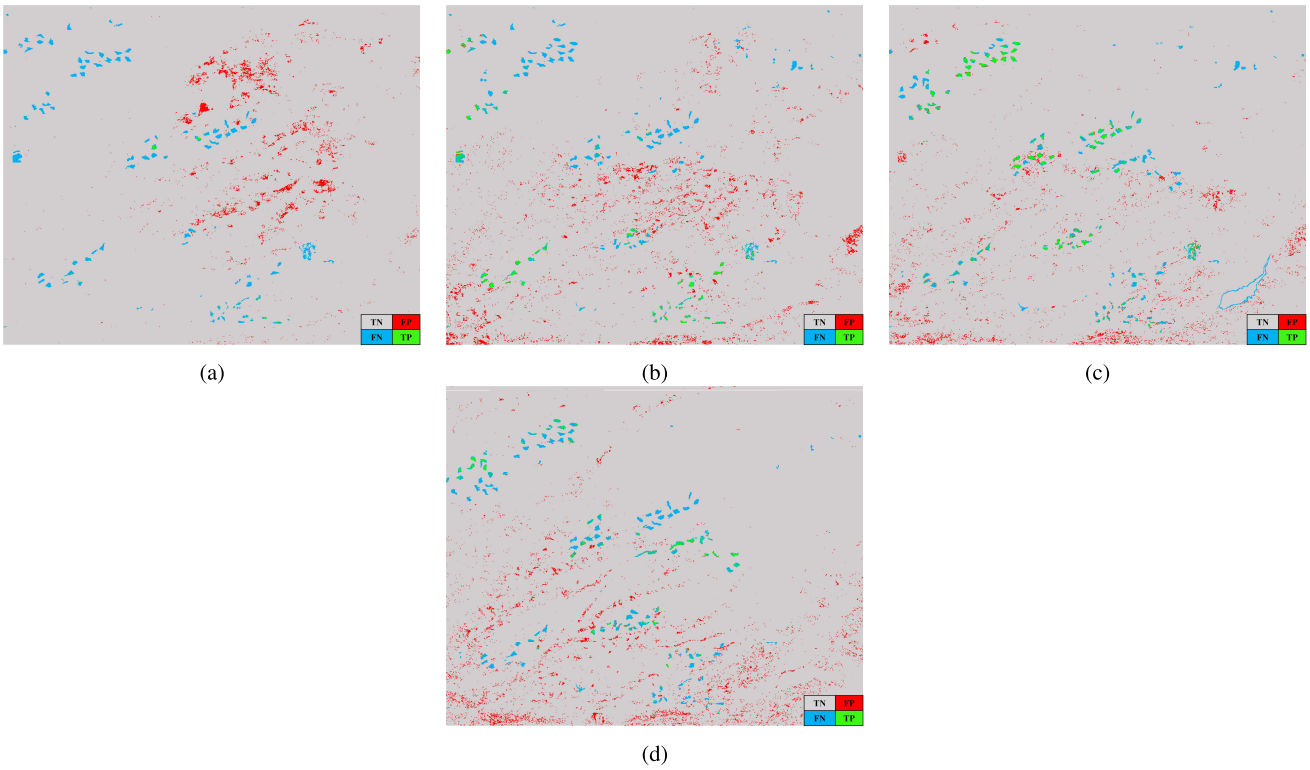


Fig. 12. CD maps of BFAST at times t = (a) 1, (b) 2, (c) 3, and (d) 4, respectively (California dataset).

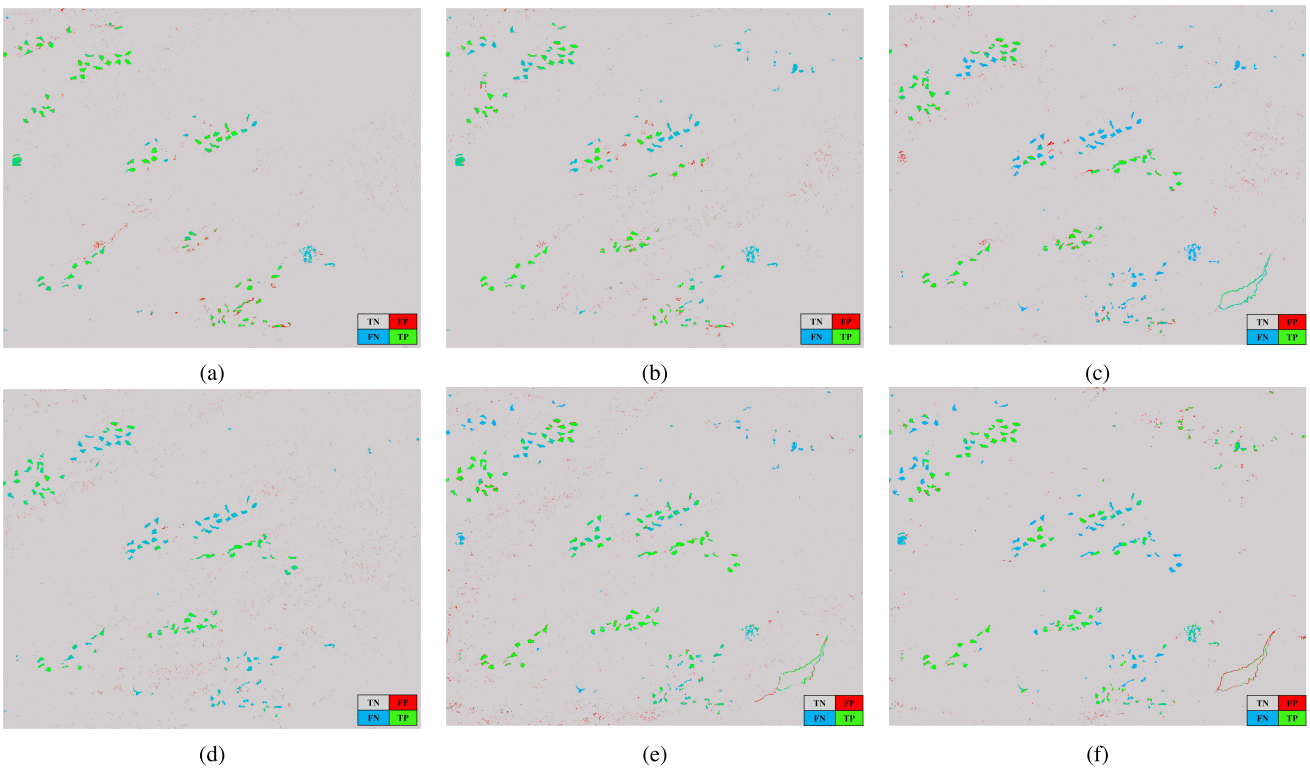


Fig. 13. CD maps of (a)–(d) represent the temporal cluster R_{t3}, each projected across the corresponding composite within the time series at times $t = 1, 2, 3$, and 4, respectively. In addition, (e) and (f) depict the results of supervised and unsupervised CVA, respectively (California dataset).

using two image sets: one from predetermined dates and the other from random dates. Comparing the CD map results of TSCVA (Figs. 11), BFAST [Fig. 12(a)–(d)], the TSCVA temporal cluster R_{t3} in 13(a)–(d) and CVA [Fig. 13(e)–(f)] it is evident that employing a specific reference vector in TSCVA allows identification of the time of deforestation and changes in lake surface. In contrast, CVA only detects changes between two images and does not facilitate differentiation between var-

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TABLE VI

FP, FN, OE, AND $F1$ SCORE METRICS FOR TEMPORALLY FOCUSED TSCVA PROJECTED AT EACH SEASONAL COMPOSITE, BFAST RESULTS, CVA, AND SPECTRALLY FOCUSED TSCVA PROJECTED AT b_2, b_4, AND b_{12}, RESPECTIVELY (CALIFORNIA DATASET)

Method	FP	FN	OE	F1
Temporal TSCVA (t1)	6757	6827	0.370	0.630
Temporal TSCVA (t2)	6957	12376	0.523	0.538
Temporal TSCVA (t3)	5073	16222	0.585	0.520
Temporal TSCVA (t4)	6857	17370	0.694	0.387
BFAST (t1)	24198	17828	0.968	0.027
BFAST (t2)	34922	18511	0.782	0.161
BFAST (t3)	25994	18810	0.678	0.285
BFAST (t4)	37732	20644	0.824	0.130
CVA (known timing)	6701	13396	0.431	0.637
CVA (unknown timing)	6066	15148	0.478	0.609
Spectral TSCVA (b2)	54959	14667	0.457	0.333
Spectral TSCVA (b4)	17042	25383	0.790	0.240
Spectral TSCVA (b12)	21226	24921	0.775	0.237

ious types of changes. Since the time series extends from May to November, several phases of deforestation were observed. Compared to BFAST, temporal TSCVA provided a clearer depiction of the progression of the deforestation patches. Although TSCVA incorporates a semisupervised cluster selection step, it works without a priori knowledge of the timing of the change. However, it is important to emphasize that these approaches are not strictly equivalent. TSCVA relies on time series composites, whereas BFAST analyzes entire time series. This analysis validates the ability of TSCVA to identify and analyze changes over time, offering a comparative perspective on its performance against BFAST and CVA.

V. CONCLUSION

In this study, we introduced TSCVA, a semisupervised framework for pixel-based CD in multispectral SITS. By extending CVA to the temporal domain, TSCVA leverages both spectral and temporal information, enabling the characterization of change magnitude and direction in a unified spectral–temporal feature space.

Empirical evaluations on Sentinel-2 data demonstrated its effectiveness across diverse scenarios, including abrupt wildfire disturbance in Greece and progressive deforestation and lake surface changes in California. These cases highlight the framework's generalizability across different land cover types, disturbance dynamics, and spectral–temporal behaviors. TSCVA provides results comparable to bitemporal CVA when exact change timings are known, but unlike CVA, it eliminates the reliance on supervised image selection and exploits the richness of time-series observations.

While our evaluation focused on medium-scale sites with relatively abrupt changes, the pixel-based design of TSCVA is inherently scalable to larger areas and transferable across ecosystems. The method is most effective for abrupt or high-magnitude changes, with reduced sensitivity to gradual or spectrally subtle variations. Future research will address these limitations by examining the role of temporal density and longer time spans, developing adaptive thresholding strategies to improve sensitivity, and removing the need for semisupervised cluster selection to achieve a fully unsupervised framework. Another promising direction is the extension of TSCVA to Very High Resolution imagery, incorporating spatial correlations through object- or region-based analysis to overcome the constraints of purely pixel-based approaches.

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REFERENCES

[1] S. Liu, Q. Du, X. Tong, A. Samat, L. Bruzzone, and F. Bovolo, "Multiscale morphological compressed change vector analysis for unsupervised multiple change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 9, pp. 4124–4137, Sep. 2017.

[2] L. Bruzzone and F. Bovolo, "A novel framework for the design of change-detection systems for very-high-resolution remote sensing images," *Proc. IEEE*, vol. 101, no. 3, pp. 609–630, Mar. 2013.

[3] A. Singh, "Review article digital change detection techniques using remotely-sensed data," *Int. J. Remote Sens.*, vol. 10, no. 6, pp. 989–1003, Jun. 1989.

[4] S. Liu, L. Bruzzone, F. Bovolo, M. Zanetti, and P. Du, "Sequential spectral change vector analysis for iteratively discovering and detecting multiple changes in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 8, pp. 4363–4378, Aug. 2015.

[5] P. Coppin, I. Jonckheere, K. Nackaerts, B. Muys, and E. Lambin, "Review ArticleDigital change detection methods in ecosystem monitoring: A review," *Int. J. Remote Sens.*, vol. 25, no. 9, pp. 1565–1596, May 2004.

[6] D. Lu, P. W. Mausel, E. S. Brondízio, and E. F. Moran, "Change detection techniques," *Int. J. remote Sens.*, vol. 25, no. 12, pp. 2365–2401, 2004.

[7] S. Kumar, M. Anouncia, S. Johnson, A. Agarwal, and P. Dwivedi, "Agriculture change detection model using remote sensing images and GIS: Study area Vellore," in *Proc. Int. Conf. Radar, Commun. Comput. (ICRCC)*, Dec. 2012, pp. 54–57.

[8] P. Du, S. Liu, P. Gamba, K. Tan, and J. Xia, "Fusion of difference images for change detection over urban areas," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 4, pp. 1076–1086, Aug. 2012.

[9] F. Bovolo and L. Bruzzone, "A split-based approach to unsupervised change detection in large-size multitemporal images: Application to tsunami-damage assessment," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 6, pp. 1658–1670, Jun. 2007.

[10] L. Alonso, J. Picos, and J. Armesto, "Automatic forest change detection through a bi-annual time series of satellite imagery: Toward production of an integrated land cover map," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 118, Apr. 2023, Art. no. 103289.

[11] J. Inglada, A. Vincent, M. Arias, B. Tardy, D. Morin, and I. Rodes, "Operational high resolution land cover map production at the country scale using satellite image time series," *Remote Sens.*, vol. 9, no. 1, p. 95, Jan. 2017.

[12] F. Bovolo, S. Marchesi, and L. Bruzzone, "A framework for automatic and unsupervised detection of multiple changes in multitemporal images," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 6, pp. 2196–2212, Jun. 2012.

[13] D. Liu and S. Cai, "A spatial–temporal modeling approach to reconstructing land-cover change trajectories from multi-temporal satellite imagery," *Ann. Assoc. Amer. Geographers*, vol. 102, no. 6, pp. 1329–1347, Nov. 2012.

[14] Z. Zhu and C. E. Woodcock, "Continuous change detection and classification of land cover using all available Landsat data," *Remote Sens. Environ.*, vol. 144, pp. 152–171, Mar. 2014.

[15] P. Arévalo, E. L. Bullock, C. E. Woodcock, and P. Olofsson, "A suite of tools for continuous land change monitoring in Google Earth Engine," *Frontiers Climate*, vol. 2, Dec. 2020, Art. no. 576740, doi: 10.3389/fclim.2020.576740.

[16] J. Verbesselt, R. Hyndman, G. Newnham, and D. Culvenor, "Detecting trend and seasonal changes in satellite image time series," *Remote Sens. Environ.*, vol. 114, no. 1, pp. 106–115, Jan. 2010.

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[17] J. Verbesselt, A. Zeileis, and M. Herold, "Near real-time disturbance detection using satellite image time series," *Remote Sens. Environ.*, vol. 123, pp. 98–108, Aug. 2012. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0034425712001150

[18] K. Zhao et al., "Detecting change-point, trend, and seasonality in satellite time series data to track abrupt changes and nonlinear dynamics: A Bayesian ensemble algorithm," *Remote Sens. Environ.*, vol. 232, Oct. 2019, Art. no. 111181. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0034425719301853

[19] H. He, J. Yan, D. Liang, Z. Sun, J. Li, and L. Wang, "Time-series land cover change detection using deep learning-based temporal semantic segmentation," *Remote Sens. Environ.*, vol. 305, May 2024, Art. no. 114101. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0034425724001123

[20] H. Zhang, Y. Teng, H. Li, and Z. Wang, "STRobustNet: Efficient change detection via spatial–temporal robust representations in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5612215, doi: 10.1109/TGRS.2025.3540794.

[21] E. Kalinicheva, D. Ienco, J. Sublime, and M. Trocan, "Unsupervised change detection analysis in satellite image time series using deep learning combined with graph-based approaches," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1450–1466, 2020.

[22] W. A. Malila, "Change vector analysis: An approach for detecting forest changes with Landsat," in *Proc. LARS Symposia*, 1980, p. 385.

[23] E. F. Lambin and A. H. Strahlers, "Change-vector analysis in multitemporal space: A tool to detect and categorize land-cover change processes using high temporal-resolution satellite data," *Remote Sens. Environ.*, vol. 48, no. 2, pp. 231–244, May 1994.

[24] F. Bovolo and L. Bruzzone, "A theoretical framework for unsupervised change detection based on change vector analysis in the polar domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 1, pp. 218–236, Jan. 2007.

[25] F. Bovolo and L. Bruzzone, "An adaptive thresholding approach to multiple-change detection in multispectral images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2011, pp. 233–236.

[26] L. Bruzzone and D. F. Prieto, "Automatic analysis of the difference image for unsupervised change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 3, pp. 1171–1182, May 2000.

[27] L. Bruzzone and D. F. Prieto, "A minimum-cost thresholding technique for unsupervised change detection," *Int. J. Remote Sens.*, vol. 21, no. 18, pp. 3539–3544, Jan. 2000.

[28] J. Chen, P. Gong, C. He, R. Pu, and P. Shi, "Land-use/land-cover change detection using improved change-vector analysis," *Photogramm. Eng. Remote Sens.*, vol. 69, no. 4, pp. 369–379, Apr. 2003.

[29] S. Singh and R. Talwar, "A comparative study on change vector analysis based change detection techniques," *Sadhana*, vol. 39, no. 6, pp. 1311–1331, Dec. 2014.

[30] S. Lhermitte, J. Verbesselt, W. W. Verstraeten, and P. Coppin, "A comparison of time series similarity measures for classification and change detection of ecosystem dynamics," *Remote Sens. Environ.*, vol. 115, no. 12, pp. 3129–3152, Dec. 2011.

[31] B. Martínez and M. A. Gilabert, "Vegetation dynamics from NDVI time series analysis using the wavelet transform," *Remote Sens. Environ.*, vol. 113, no. 9, pp. 1823–1842, Sep. 2009.

[32] A. Jacquin, D. Sheeren, and J.-P. Lacombe, "Vegetation cover degradation assessment in madagascar savanna based on trend analysis of MODIS NDVI time series," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 12, pp. S3–S10, Feb. 2010.

[33] Y. Lin, L. Zhang, and N. Wang, "A new time series change detection method for Landsat land use and land cover change," in *Proc. 10th Int. Workshop Anal. Multitemporal Remote Sens. Images (MultiTemp)*, Aug. 2019, pp. 1–4.

[34] W. C. Olding, J. C. Olivier, B. P. Salmon, and W. Kleynhans, "A forecasting approach to online change detection in land cover time series," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 5, pp. 1451–1460, May 2019.

[35] H. Yin, D. Pflugmacher, R. E. Kennedy, D. Sulla-Menashe, and P. Hostert, "Mapping annual land use and land cover changes using MODIS time series," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 8, pp. 3421–3427, Aug. 2014.

[36] S. Saha, F. Bovolo, and L. Bruzzone, "Change detection in image time-series using unsupervised LSTM," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[37] J. Yan, L. Wang, W. Song, Y. Chen, X. Chen, and Z. Deng, "A time-series classification approach based on change detection for rapid land cover mapping," *ISPRS J. Photogramm. Remote Sens.*, vol. 158, pp. 249–262, Dec. 2019.

[38] K. Meshkini, F. Bovolo, and L. Bruzzone, "An unsupervised change detection approach for dense satellite image time series using 3D CNN," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 4336–4339.

[39] B. P. Salmon, J. C. Olivier, K. J. Wessels, W. Kleynhans, F. van den Bergh, and K. C. Steenkamp, "Unsupervised land cover change detection: Meaningful sequential time series analysis," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 4, no. 2, pp. 327–335, Jun. 2011.

[40] V. Ortiz-Rivera, M. Vélez-Reyes, and B. Roysam, "Change detection in hyperspectral imagery using temporal principal components," *Proc. SPIE*, vol. 6233, pp. 368–377, Mar. 2006.

[41] R. J. Radke, S. Andra, O. Al-Kofahi, and B. Roysam, "Image change detection algorithms: A systematic survey," *IEEE Trans. Image Process.*, vol. 14, no. 3, pp. 294–307, Mar. 2005.

[42] M. Zanetti, F. Bovolo, and L. Bruzzone, "Rayleigh-rice mixture parameter estimation via EM algorithm for change detection in multispectral images," *IEEE Trans. Image Process.*, vol. 24, no. 12, pp. 5004–5016, Dec. 2015.

[43] F. G. B. De Natale and G. Boato, "Detecting morphological filtering of binary images," *IEEE Trans. Inf. Forensics Security*, vol. 12, no. 5, pp. 1207–1217, May 2017.

[44] P. Maragos, "Morphological filtering," in *The Essential Guide to Image Processing*, A. Bovik, Ed., Boston, MA, USA: Academic, 2009, ch. 1, pp. 293–321. [Online]. Available: https://www.sciencedirect.com/science/article/pii/B9780123744579000135

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